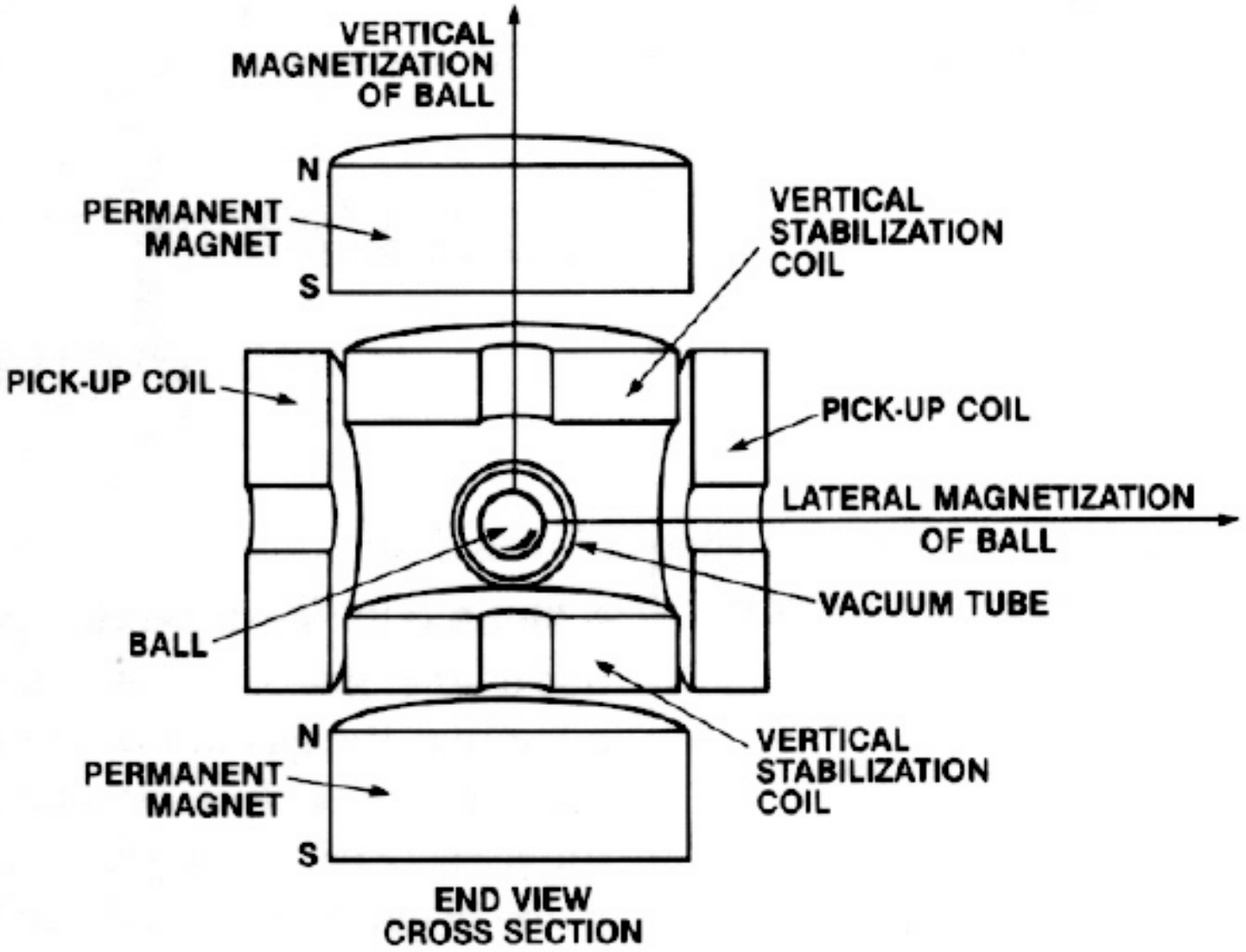
Pressure Measurement

Pressure is defined as force per unit of area: $p = F / A$,where F is force and A the area to which the force is applied.

If pressure is measured via the force that is exerted on an area, it is termed a pressure measurement that is independent of the type of gas in question. Pressure measurement on the basis of force reaches its limits at pressures of less than 1 mbar, because the exerted forces become too small.

Consequently other processes must be used. The thermal conductivity of the enclosed gas can be used, for example, or the gas molecules can be ionized, the ion streams flowing between electrodes measured.

A spinning rotor gauge (SRG), a so-called gas friction gauge, is used for calibration purposes. A sphere is magnetically suspended in the vacuum and caused to rotate rapidly, at which point the drive is then de-energized. The pressure of the type of gas that is present can be calculated from the decrease in rotational frequency due to gas friction. These devices measure pressures p of more than 10^{-7} mbar. Calibrated spheres can be used as a transfer standard.

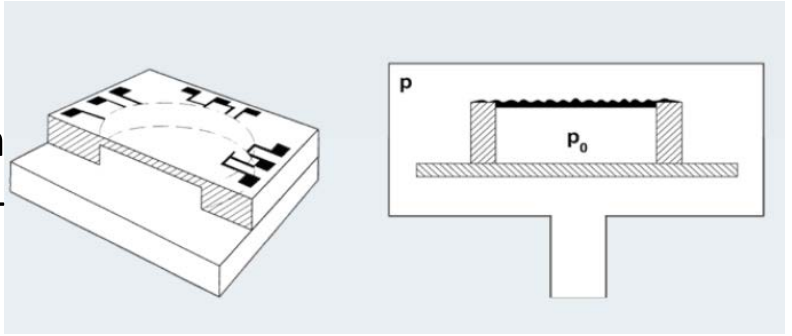


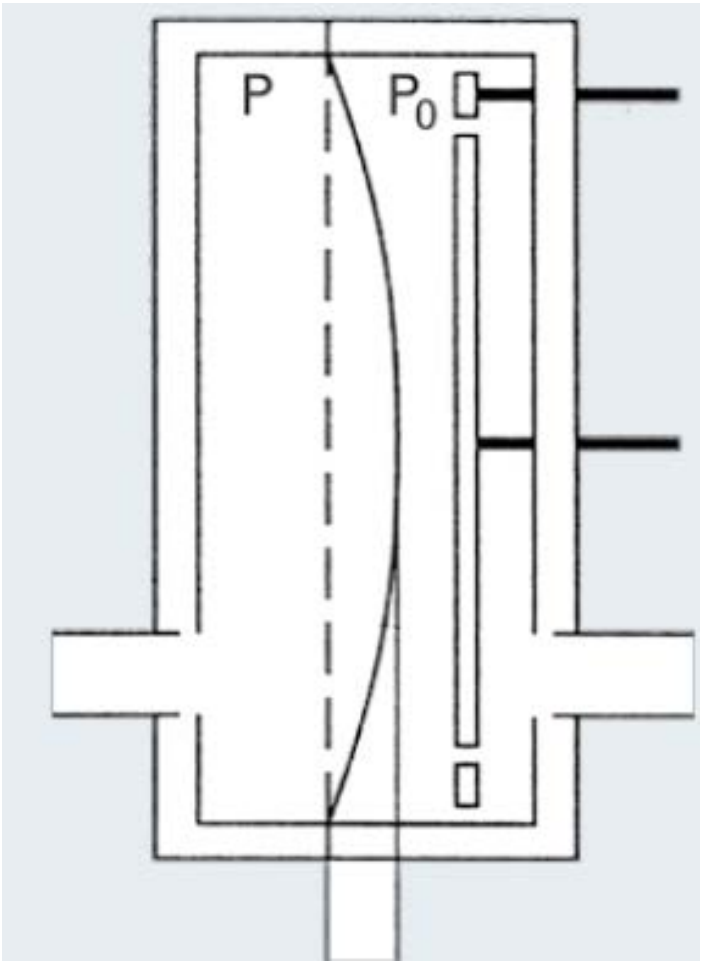
Primary calibration - rotating ball gauge

Direct Gauges

Piezo-diaphragm vacuum gauges

A simple and extremely robust method involves the use of a piezo-resistive pick-up. The design is shown in Figure. A diaphragm is arranged over a well-evacuated volume having a reference pressure *p0 , into which the expansion measurement resistances are diffused.* The measured change in resistance as a result of diaphragm deflection serves as a parameter for the pressure. This pick-up is characterized by its insensitivity to gas inrush and its high accuracy.





Capacitive diaphragm vacuum gauges

In a capacitive vacuum gauge, deflection of the diaphragm is measured as the change in capacity of a plate capacitor that is formed by the diaphragm and a fixed counter plate in a well-evacuated space having a pressure *p0. The diaphragm is comprised either of* ceramic with a vacuum-metalized coating or of stainless steel. The lower measurement limit is 10^{-5} mbar.

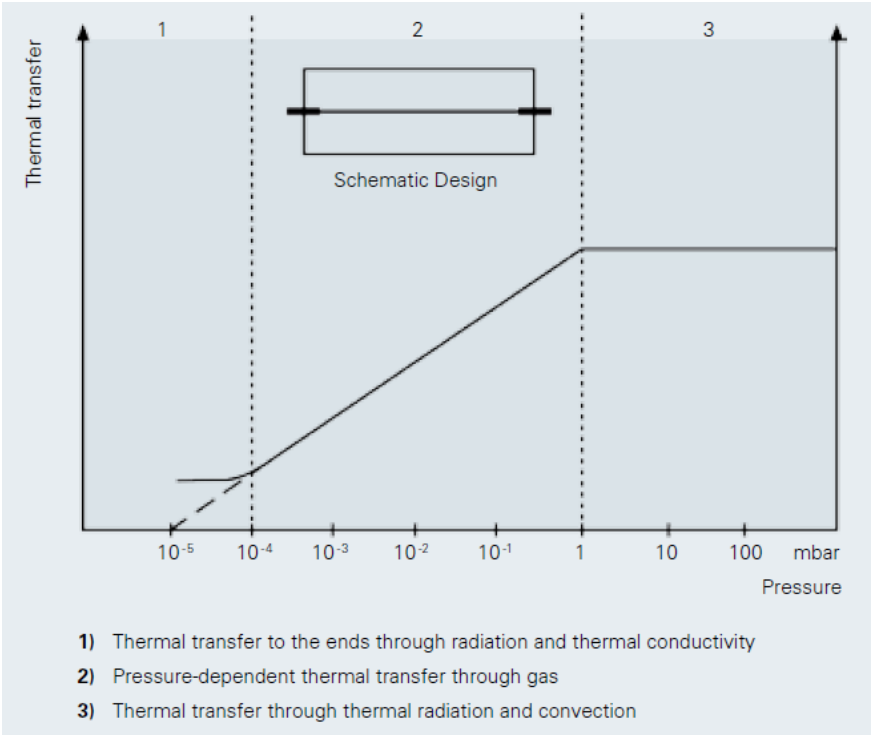
Indirect, gas-dependent pressure measurement

•At extremely low pressures, the influence of the force on a diaphragm becoming negligible.

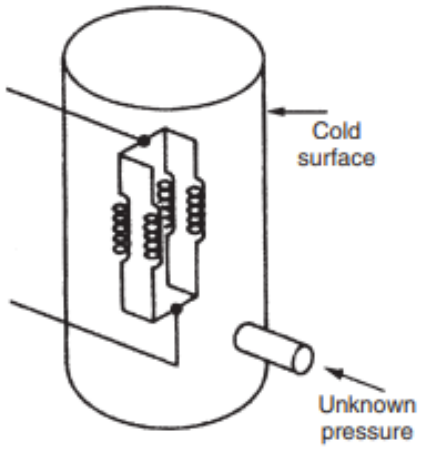
•This is why pressure is determined by means of the molecular number density, which is proportional to pressure. The status equation that applies for an ideal gas is: $p = n \cdot k \cdot T$. Thus, pressure is proportional to molecular number density where temperature *T is identical. This formula is satisfied for the pressures that prevail in vacuum technology.*

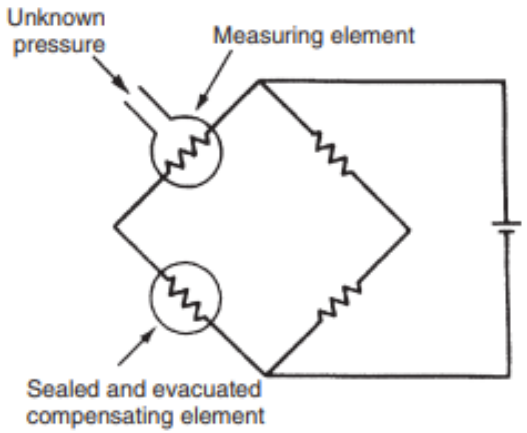
•Various physical effects, such as thermal transfer, ionization capacity or electrical conductivity, are measured for this purpose. These values are a function of both pressure as well as molecular weight. This results in a pressure measurement that produces differing results for different heavy gases.

Pirani vacuum gauge



1) Thermal transfer to the ends through radiation and thermal conductivity
2) Pressure-dependent thermal transfer through gas
3) Thermal transfer through thermal radiation and convection





A Pirani vacuum gauge utilizes the thermal conductivity of gases at pressures *p of less than* 1 mbar. Wire (usually tungsten) that is tensioned concentrically within a tube is electrically heated to a constant temperature between 110 °C und 130 °C by passing a current through the wire. The surrounding gas dissipates the heat to the wall of the tube. In the molecular flow range, the thermal transfer is the molecular number density and is thus proportional to pressure. If the temperature of the wire is kept constant, its heat output will be a function of pressure. However it will not be a linear function of pressure, as thermal conductivity via the suspension of the wire and thermal radiation will also influence the heat output.

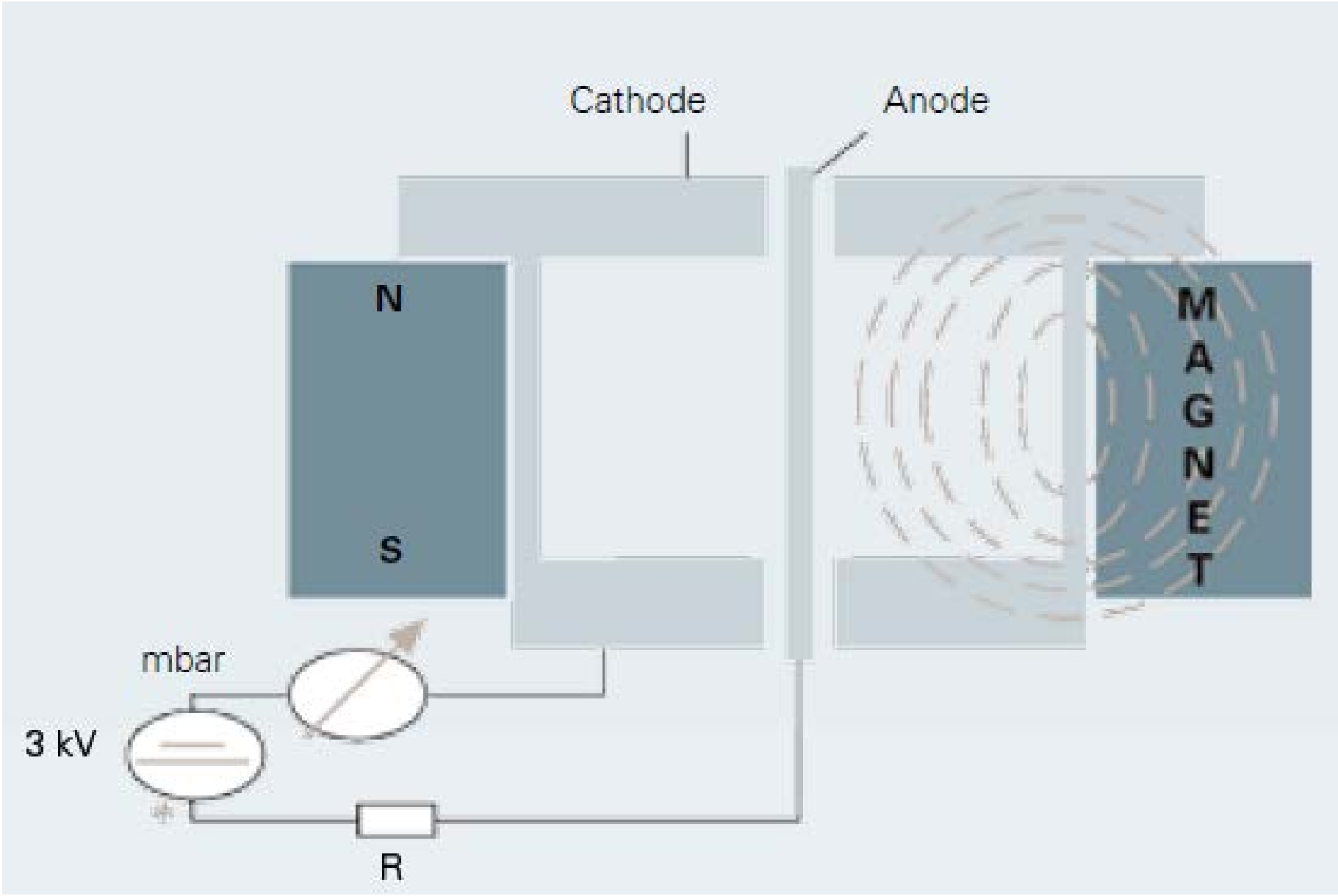
The limiting effects are:

•Thermal conductivity will not be a function of pressure in the range of 1 mbar to atmospheric pressure (laminar flow range)

•The thermal conductivity of the gas will be low relative to the thermal transfer over the wire ends at pressures below 10^{-4} mbar, and will thus no longer influence the heat output of the wire. Consequently, the measurement limit is approximately at 10^{-4} mbar.

•Thermal radiation will also transfer a portion of the heat output to the wall of the tube

Cold cathode ionization vacuum gauges



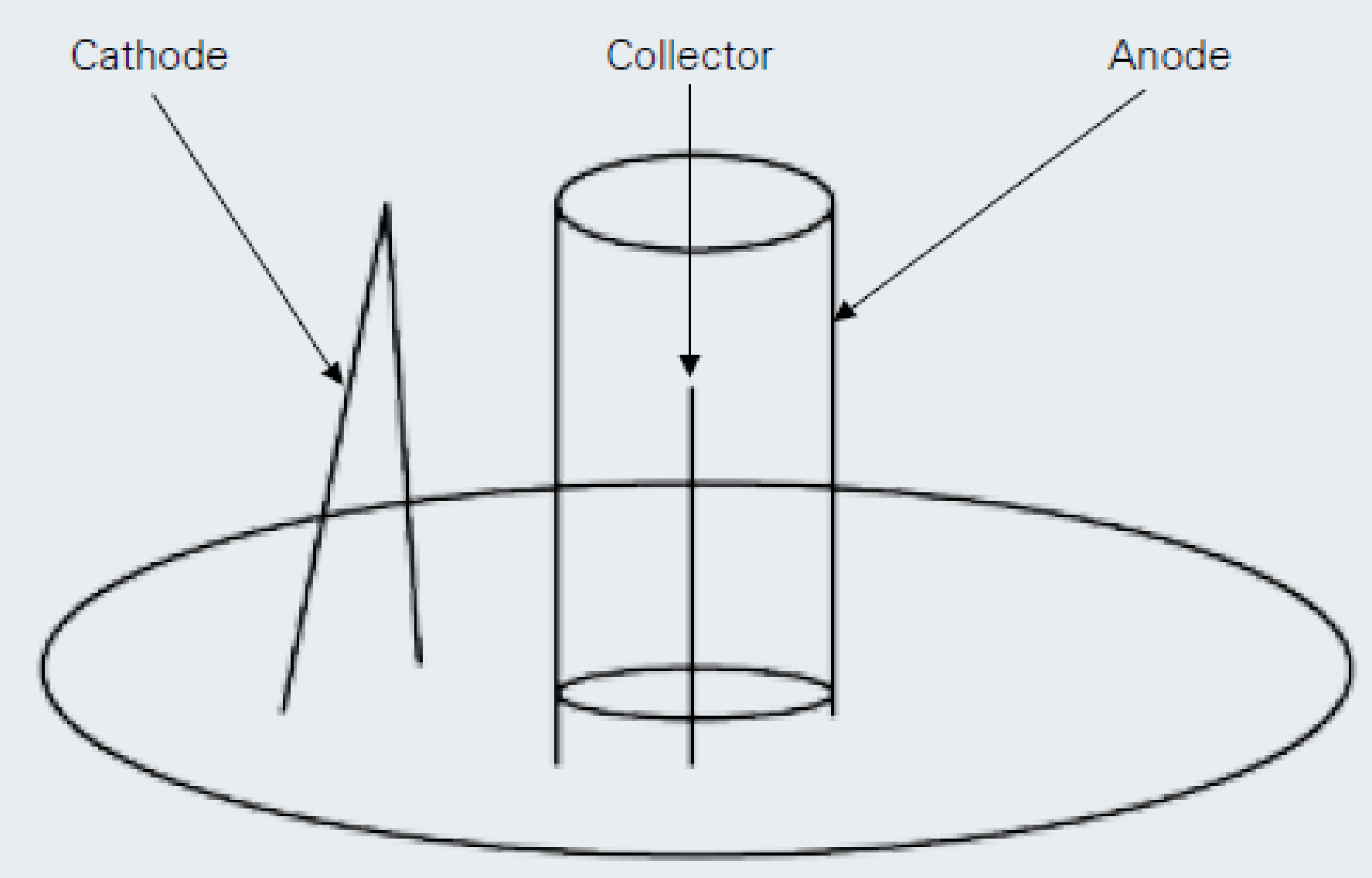
Cold cathode ionization vacuum gauges

- Cold cathode ionization vacuum gauges essentially consist of only two electrodes, a cathode and an anode, between which a high voltage is applied via a series resistor. Negatively charged electrons leave the cathode because of the high voltage, moving at high velocity from the cathode toward the anode. As they travel this path, they ionize neutral gas molecules, which ignites a gas discharge. The measured gas discharge current is a parameter for pressure. However only few molecules are ionized with straight electron trajectories, which results in lower sensitivity and interruption of the gas discharge at approximately 1 mbar.

- A design that avoids this disadvantage is the inverted magnetron after Hobsen and Redhead. A metal pin (anode) is surrounded by a rotationally symmetrical measurement chamber (cathode). An axially magnetized, cylindrical, permanent-magnet ring is placed on the exterior of the measurement chamber to generate a magnetic field within the chamber.

- The electrons travel through the magnetic field on spiral trajectories. The electron paths extended in this manner increase the probability of collisions with the gas molecules and ensure that sufficient ions are generated to maintain the gas discharge, even at pressures of less than 1 mbar. The pressure reading will depend upon the type of gas in question due to the different ionization probabilities of the various gases.

Hot cathode ionization vacuum gauges



Hot cathode ionization vacuum gauges

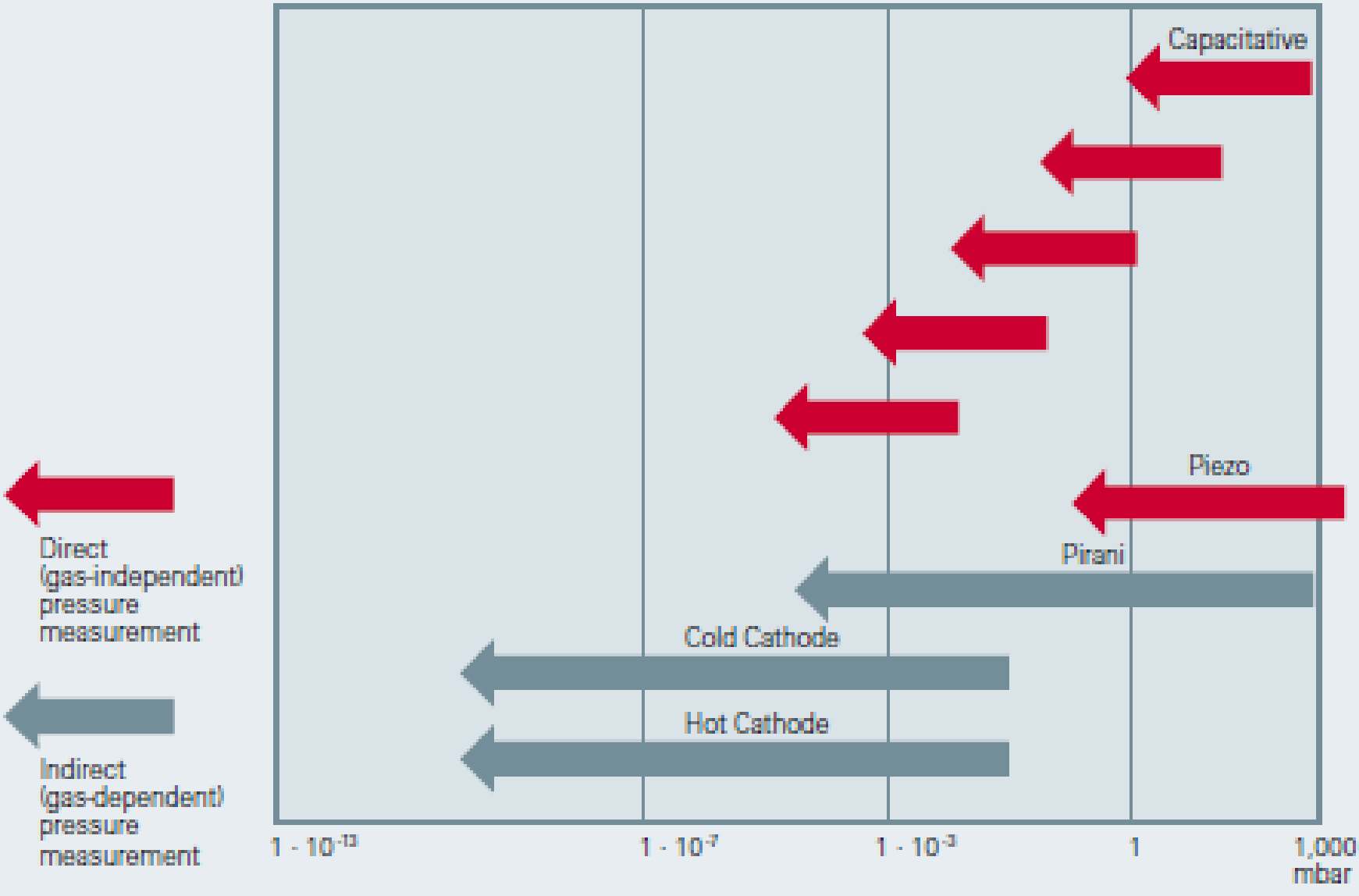
In this case, as opposed to cold cathode ionization vacuum gauges, electrons are generated with the aid of a heated cathode. A thin wire is arranged in the middle of the cylindrical, lattice-shaped anode; this wire serves as the ion collector. A voltage of approximately 100 V is applied between anode and cathode. This accelerates all emitted electrons toward the anode. The emission current is measured in the anode circuit, which can be set by means of the heat output of the cathode. As they travel toward the anode, gas molecules that strike the collector, which has the potential of the cathode, are ionized by electron collisions. The measured collector current is a parameter for pressure. Since the emission current is proportional to the ion current, it can be used to set the sensitivity of the gauges. Pressures can be accurately measured to $1 \cdot 10^{-10}$ mbar with Bayard-Alpert sensors.

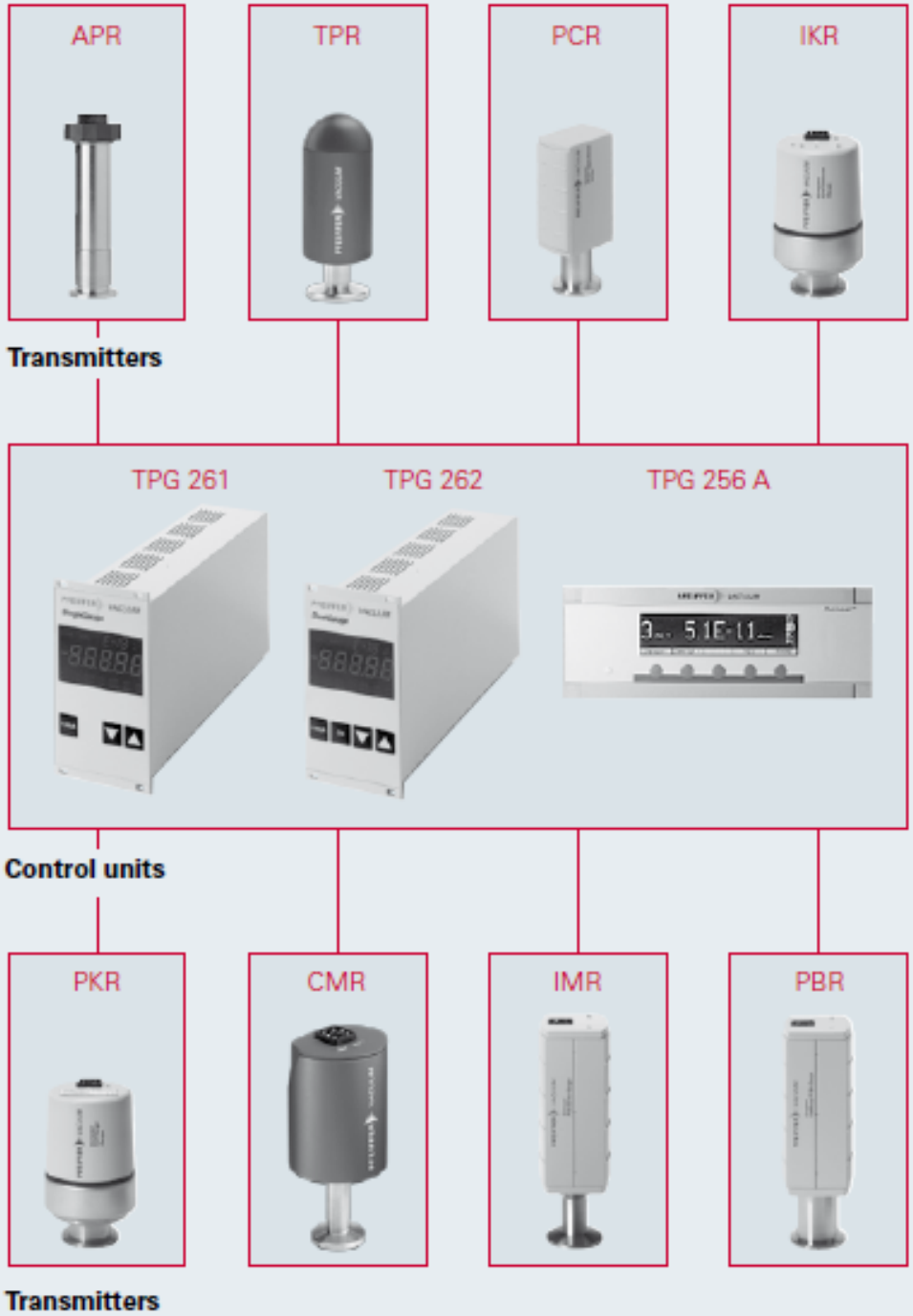
Limits:

X-ray braking radiation Electrons that strike the anode cage cause x-rays to be emitted, some of which strike the collector. This x-ray effect causes the collector to emit photoelectrons that flow off toward the anode. The resulting photoelectron current increases and falsifies the pressure-dependent collector current. Consequently the collector wire should be selected as thin as possible so that it collects only little x-ray radiation.

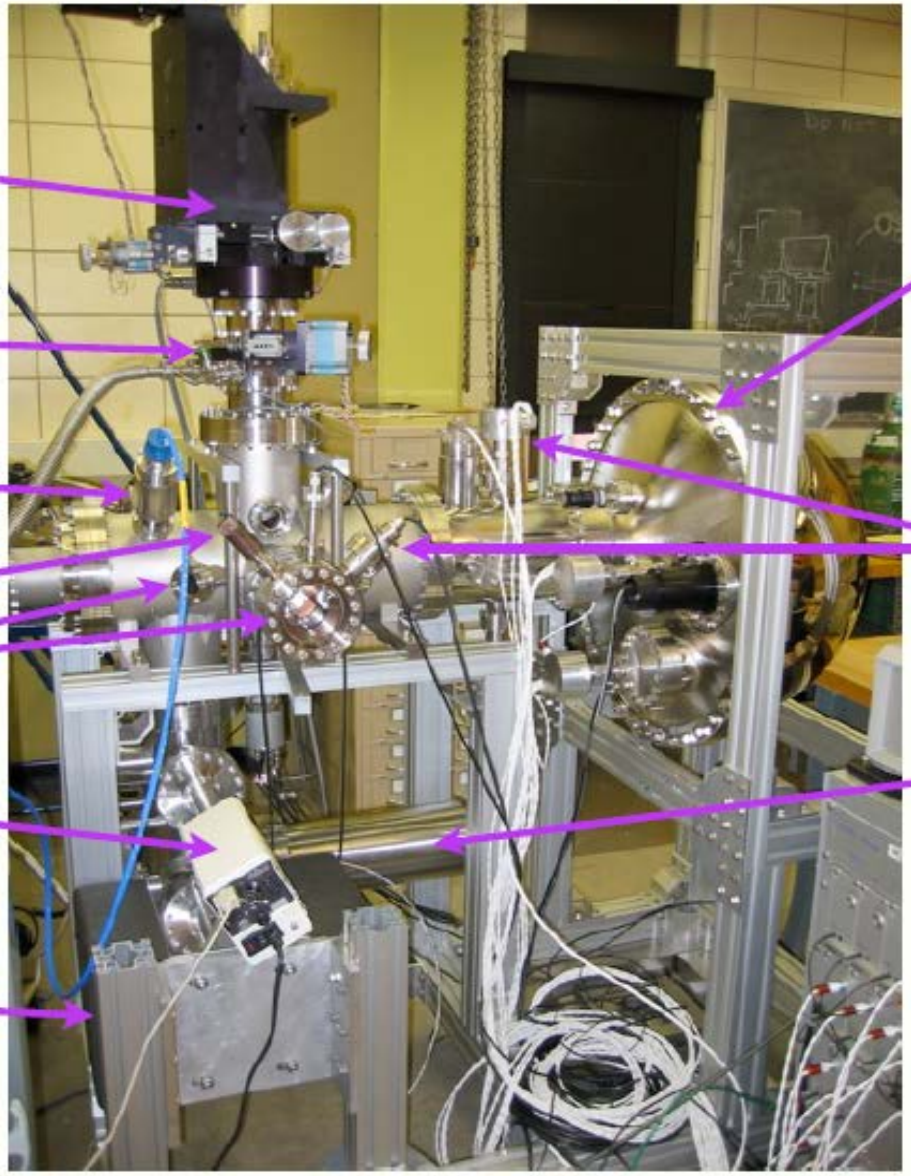
ESD ions ESD (electron stimulated desorption) means that gas molecules deposited on the anode cage are desorbed and ionized by electrons. These ions also increase the pressure-proportional ion current.

Vacuum gauges operational ranges




APR
TPR
PCR
IKR
Transmitters
TPG 261
TPG 262
TPG 256 A
Control units
PKR
CMR
IMR
PBR
Transmitters

Typical UHV system



XYZ manipulator

Rotary seal

Ion gauge

Rotary feedthru

Viewports

RGA
(residual gas analyzer)

Ion pump

Really big UHV flange

Electrical feedthru's

TSP
(Titanium sublimation pump)

Chambers



Stainless steel 304, 316:
easy to machine/weld
but transparent to H2
$p > 10^{-11}$ Tr

Aluminum:
fragile, expensive to make
but resistant to H2
$p < 10^{-14}$ Tr

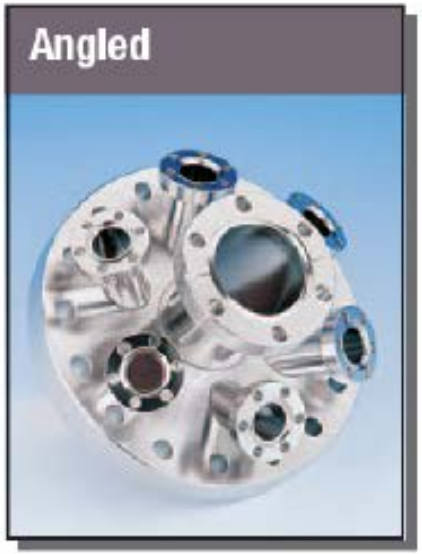
Components



)0° Del-Seal CF Elbows page 66



Del-Seal CF Reducing Cross page 71







)el-Seal CF 5-Way Cross page 72

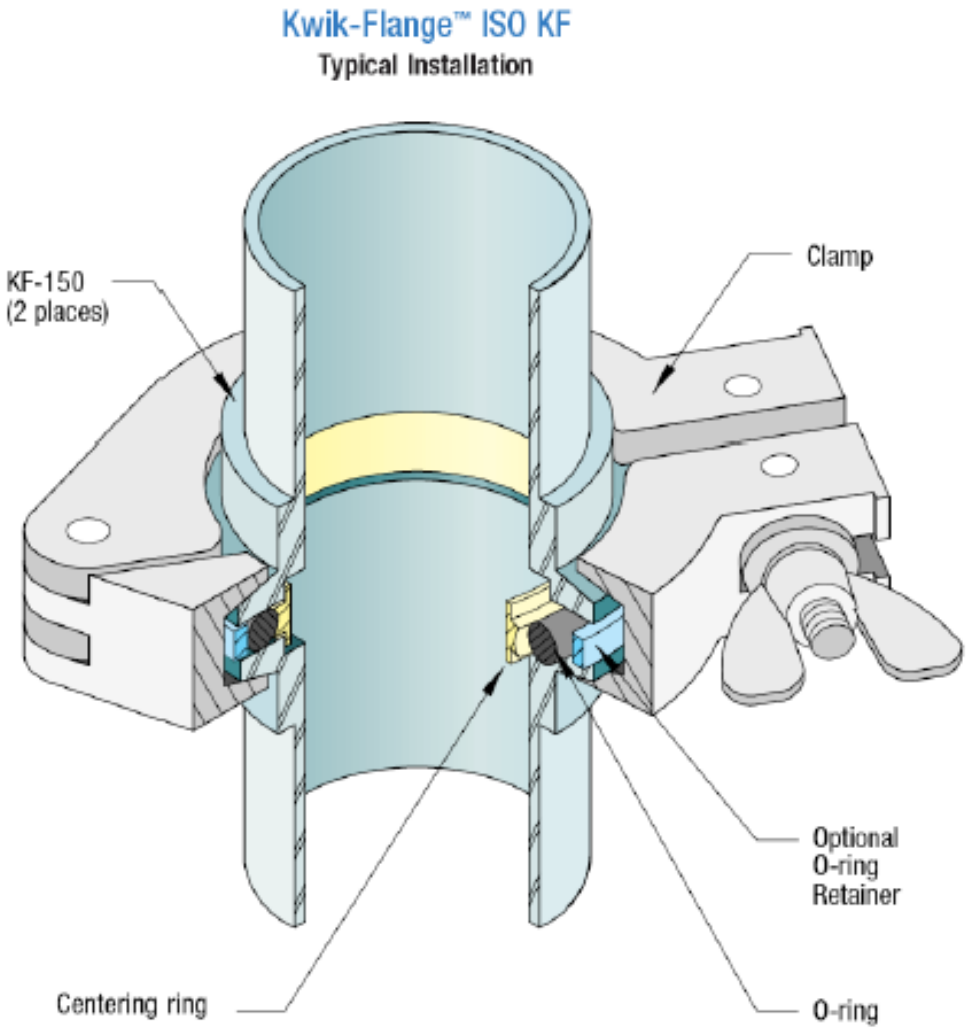


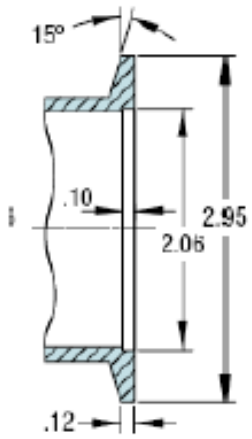
Del-Seal CF 6-Way Cube page 73

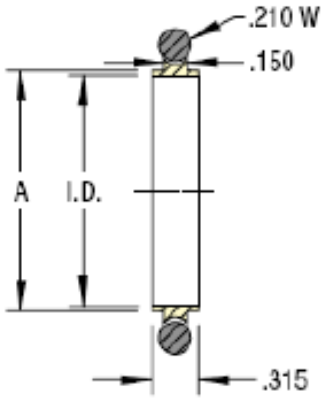




Kwik flange ISO KF

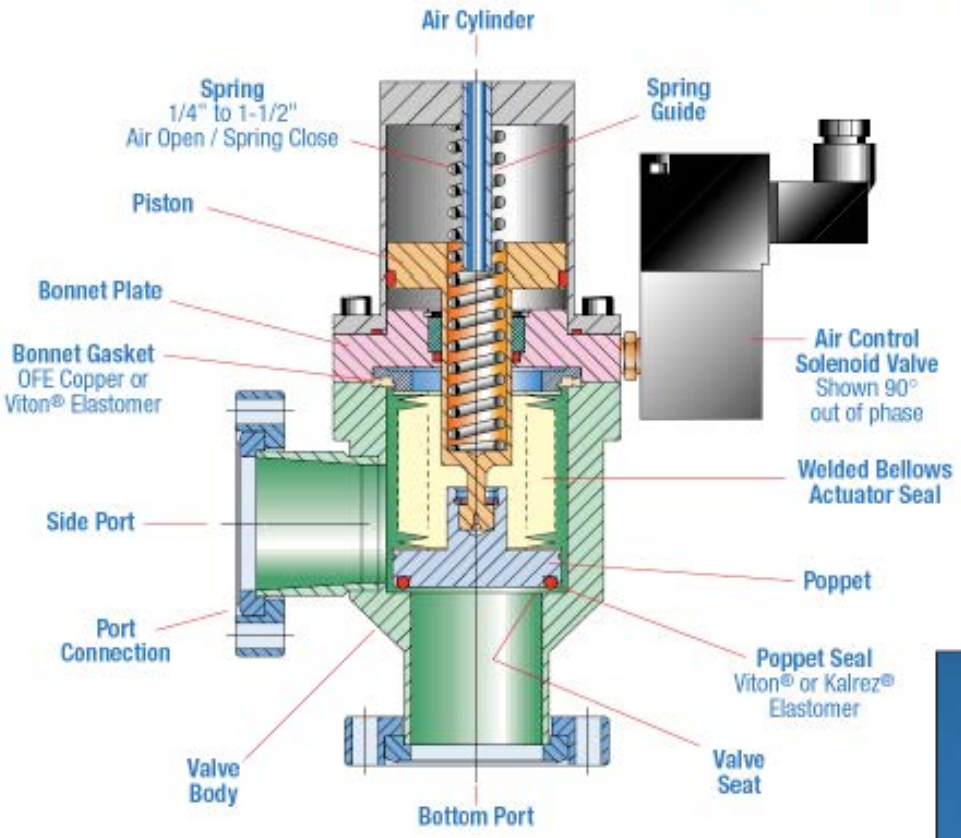


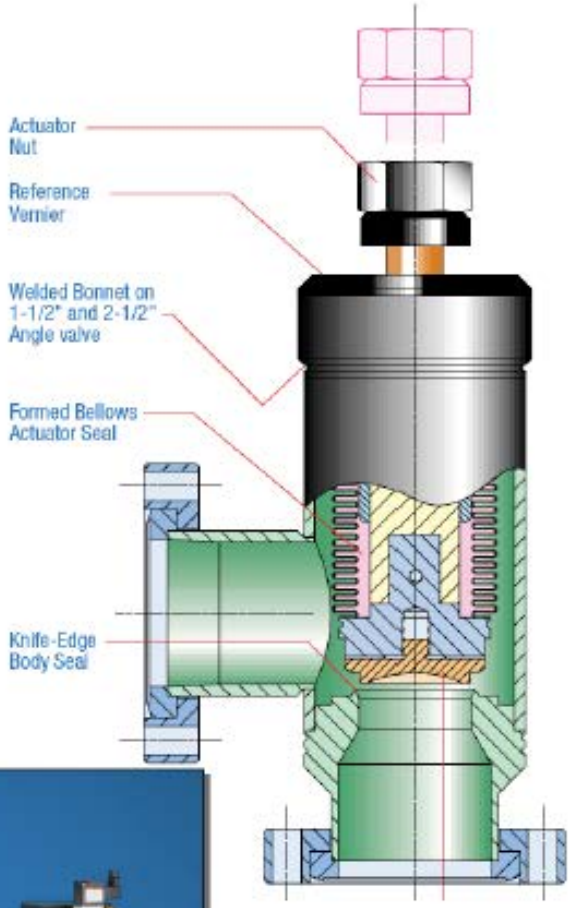






90° valves


Air Cylinder
Spring
1/4" to 1-1/2"
Air Open / Spring Close
Spring
Guide
Piston
Bonnet Plate
Bonnet Gasket
OFE Copper or
Viton® Elastomer
Air Control
Solenoid Valve
Shown 90°
out of phase
Welded Bellows
Actuator Seal
Side Port
Poppet
Port
Connection
Poppet Seal
Viton® or Kalrez®
Elastomer
Valve
Body
Valve
Seat
Bottom Port


Actuator
Nut
Reference
Vernier
Welded Bonnet on
1-1/2" and 2-1/2"
Angle valve
Formed Bellows
Actuator Seal
Knife-Edge
Body Seal


AV-600
AV-600-PAA

Gate valves



Del-Seal™ CF



10" Del-Seal CF



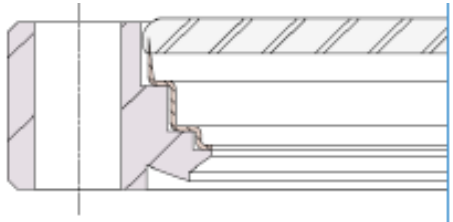
Viewports





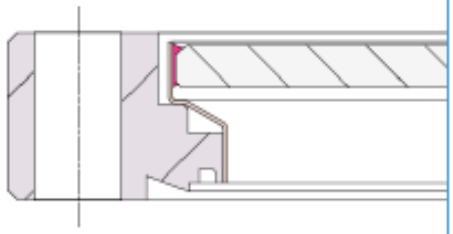
Glass

Glass is fused to a thin nickel-iron metal transition



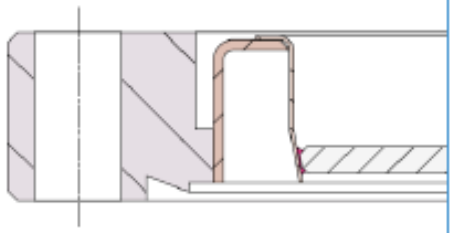
Quartz

Quartz is metalized and brazed directly to stainless steel using a lead-silver braze alloy



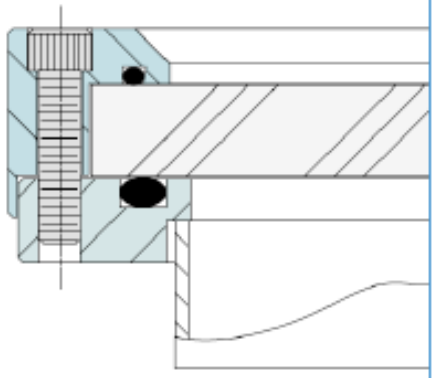
Sapphire

Sapphire is metalized and vacuum brazed to a nickel-iron sleeve using a tapered seal interface



Pyrex®

A Pyrex® substrate is sandwiched between two elastomer seals. The outer elastomer (top in drawing) is for cushioning and the inner elastomer (bottom in drawing) makes the vacuum seal.



Electrical feedthru's









Motion feedthru's

Rotary



BRM-133

Linear



BLM-133-1

Rotary +Linear



Linear Drive



PBRM2-10

Wobblestick

Blank Jaws



DG-275

Patented Design

Sample manipulation



